QUIZ#2 Sec 1

Solve the IVP

$$\begin{cases} y'' - 6y' + 9y = 4e^{3t} \\ y(0) = 1 \,, \; y'(0) = 3 \end{cases}$$

Solution

$y_h = \alpha e^{3t} + \beta t e^{3t}$, $\alpha, \beta \in \mathbb{R}$

For a y_p GUESS: at^2e^{3t}

$a[9t^2 - 6(3t^2+2t) + (9t^2+12t+2)] = 4$

$\Rightarrow 2a = 4 \Rightarrow a = 2$

So the soln must be of the form $y = 2t^2e^{3t} + \alpha e^{3t} + \beta t e^{3t}$

$$\left.\begin{aligned} y(0) &= \alpha + \beta = 1 \\ y'(0) &= 3\alpha + \beta = 3 \end{aligned}\right\} \alpha = 1, \beta = 0$$

soln: $y = 2t^2e^{3t} + 2e^{3t} - te^{3t}$

Let's change the forcing fnc:

Solve the IVP

$$\begin{cases} y'' - 6y' + 9y = 4te^{3t} \\ y(0) = 1, \quad y'(0) = 3 \end{cases}$$

For this question at^2e^{3t} does not help. You might use var of param Or easier: multiply each side by e^{-3t}, and set $u = ye^{-3t}$: $u' = y'e^{3t} - 3ye^{3t}$,

$$u'' = y''e^{-3t} - 6y'e^{-3t} + 9ye^{-3t}$$
$$= e^{-3t} \cdot L[y] = 4t$$

$$\Rightarrow u = \frac{2}{3}t^3 \Rightarrow y_p = \frac{2}{3}t^3e^{3t}$$